

Experimental and computational investigation of fault diagnosis of unbalance mass in rotating machinery

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Abstract

This study delves into investigating various aspects of machinery performance and vibration analysis through experimental simulations using the Machinery Fault Simulator (MFS). It examines the influence of different masses and speeds on machinery dynamics, considering both time and frequency domain data. This study explores the effects of rotating unbalanced mass in single and multiple planes on machinery vibration, providing insights into diagnosis and mitigation strategies. The results demonstrate how the placement of an unbalanced mass in relation to each other in single and multiple planes varies the vibration effect, highlighting the importance of proper balancing and maintenance practices for optimizing machinery performance and longevity. Experimental results are validated with the 3D Finite Element (FE) result for a few cases showing excellent agreement. In general the study underscores the importance of MFS technology in advancing fault diagnosis and predictive maintenance in industrial settings.

Keywords

Rotating unbalance mass, fault simulator, FE simulation, fault diagnosis, vibration analysis

Introduction

In quality monitoring, one of the essential parts is to detect faults in manufacturing systems management. Information about functional conditions enables operations to improve quality and increase throughput.¹ Vibration monitoring is increasingly important for maintenance because it helps predict and prevent problems, leading to more efficient operations. Vibration analysis serves as a valuable tool in predictive maintenance, enabling early fault detection and efficient resource management.² Gangsar et al. introduces an automated approach for diagnosing rotor system unbalance by employing a machine learning technique known as Support Vector Machine (SVM). The methodology involves collecting vibration signals from both balanced and unbalanced rotor conditions, extracting statistical features such as standard deviation, skewness, and kurtosis from these signals, and utilizing these features to train and test the SVM model for effective fault classification.³ Phadatare and Pratiher investigated vibrations in fast-spinning rotors with unbalanced weight considering nonlinearity in geometry and concluded that with increasing shaft speed, the response amplitude becomes significant under mass unbalance in the rotor-bearing system. However, for rotational speeds below 200 rad/s, the effect of mass unbalance was negligible.⁴ Yadav et al. examined rotor behavior with unbalanced conditions and extra space in roller bearings as

a source of nonlinearity and found that the rotor-bearing system exhibits varying dynamic behaviors, transitioning from stable and unstable second-order quasiperiodicity to third-order and chaotic responses at specific speeds, primarily influenced by unbalanced forces.⁵ Rahman and Uddin used vibration signals to find faults in induction motors using time and frequency domain analysis of current,⁶ while Song et al. automated the diagnosis of low-speed bearings through vibration analysis by using filters for canceling the noise and then collecting vibration signal.⁷ Early detection of these issues is crucial for extending machine life and reducing maintenance costs. Therefore, researchers are exploring how misalignment affects machines through experiments and simulations. Other researchers, like Khaire and Khot conducted experimental studies to examine the impact of misalignment on rotor-bearing systems.⁸ Additionally, Khaire and Phalle utilized machine learning models to detect various bearing failures at different

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operating speeds.⁹ Different vibration analysis techniques are used, specifically, in the time, frequency, and time-frequency domains, to analyze signals related to faults in rolling element ball bearings. Vibration spectra of healthy bearings, as well as those with inner and outer race defects, are processed using the fast Fourier transformation (FFT) and the inverse fast Fourier transformation (IFFT).¹⁰ The cracked shaft was analyzed for longitudinal and bending vibration and found the location of the crack in the rotating shaft.¹¹ Various faults like unbalanced mass, bent shaft, misalignment were also analyzed for vibration monitoring by Patil et al.¹² Sai and Wang gave a diagnostic method to determine the position and size of transverse crack in static and rotating shaft.¹³ Chen et al aimed to summarize and review the latest research and development trends in the fundamental area of machinery fault diagnosis, focusing on four key aspects: fault mechanisms, sensor technology and signal acquisition, signal processing, and intelligent diagnostics.¹⁴ Few researchers have applied various machine learning techniques to perform fault diagnosis. The investigation of rotating machinery fault diagnosis has advanced over the years and is still active. There are two main methods, one of which is to extract the characteristics of vibration signals and then use machine learning algorithms for fault diagnosis.¹⁵ Lin et al. gave a novel method which applies two-phase parameter identification: first, bearing parameters and initial unbalances are evaluated with a flexible shaft; then, daily unbalance progression is monitored with fixed bearing parameters. This allows effective diagnosis and prognosis of rotor unbalance.¹⁶ The study uses the design of experiments method to analyze how unbalanced disks in different numbers and positions affect the vibration of a dual-rotor system. Vibration responses at various speeds and measurement points are examined to understand their relation to unbalance parameters, helping identify the system's dynamic behavior and guide aircraft rotor balancing.¹⁷ The research aims of Abbasi in 2022 was to enhance the accuracy of unbalance parameter estimation for critical engineering use. A hybrid algorithm is proposed,

with the objective function based on weighted error between analytical and predicted responses. Parameters are optimized using Taguchi DOE, and its efficiency is compared with recent algorithms.¹⁸ In general, these studies highlight the importance of early fault detection to extend machine life and reduce costs. From the above literature survey it is very clear that very few researchers have worked in studying rotating unbalance masses in multiple planes and hence this forms a research gap.

In this paper, the Tiera Machinery Fault Simulator was employed to investigate the effects of rotating unbalance in both single and multi-plane configurations on vibration behavior over time and across different frequency ranges. Vibration signals were analyzed in both the time and frequency domains to identify dominant frequencies, which were determined by locating the highest amplitude peaks. Additionally, the vibration response was examined for scenarios involving multiple rotating masses located in a single plane as well as in two separate planes, providing further insight into the influence of mass distribution on system dynamics.

The structure of this paper is organized into four additional sections. Section 2 provides a detailed explanation of Machinery Fault Simulator with Experimental Setup followed by Experimental Methodology in Section 3. The results and discussion are explained in Section 4. Finally, Section 5 elaborates on the conclusion and future scope of this work.

Machinery fault simulator

The Machinery Fault Simulator (MFS), shown in [Figure 1](#) is a reliable and affordable tool for early detection of machinery faults, helping prevent damage and production issues. Effective fault detection relies on clear analysis of vibration patterns, waveforms, and phase data. With a plant running at full production, it is virtually impossible to gain an understanding of the kinetics and dynamics of machinery without adversely affecting production and profits. The MFS enables off-line training and experimentation,

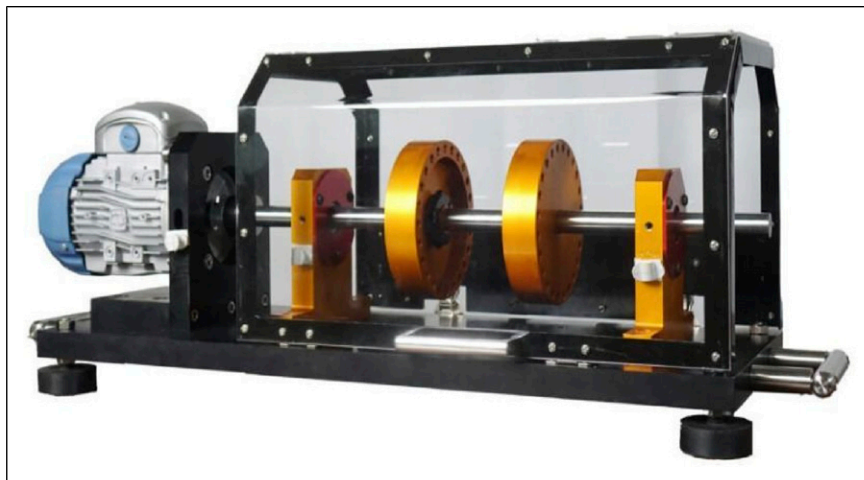


Figure 1. Tiera machinery fault simulator.¹⁹

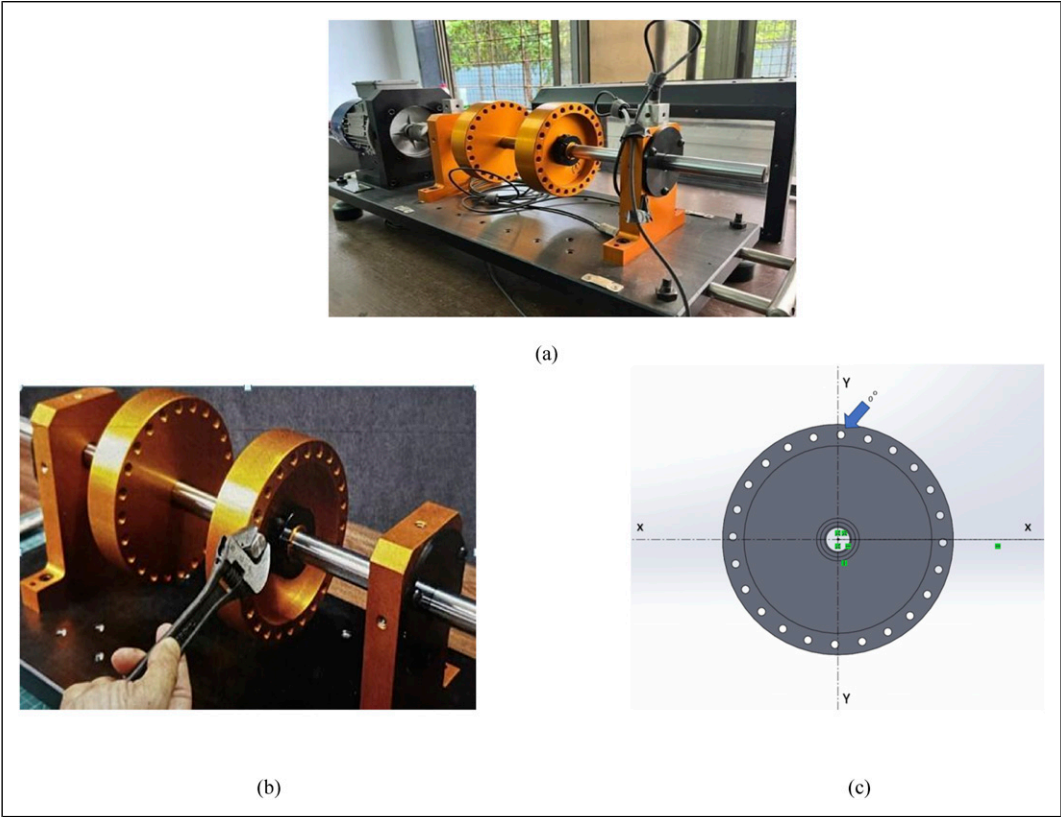


Figure 2. Rotor in single plane (a) no mass (b) mass at 30 degree (c) Rotor with x-y axis and zero degree position.

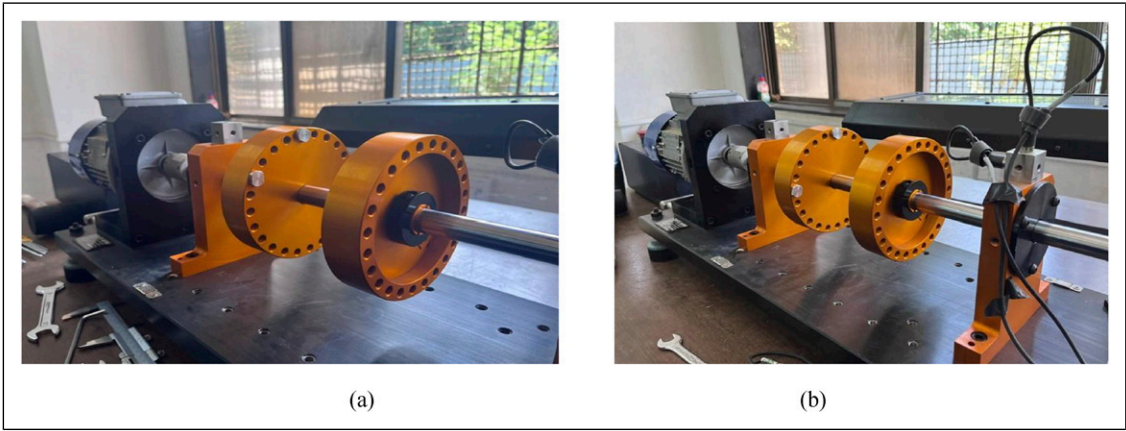


Figure 3. Rotating unbalance masses in a single plane (a) at 60 degree (b) at 90 degree.

Table 1. Trials details.

Sr. No.	Mass	Magnitude (gram)	Plane	Position (Degree)	Speed (rpm)
1	Single	1.4	Single	0	500, 1000, 1500
		1.92	Single	0	500, 1000, 1500
		2.68	Single	0	500, 1000, 1500
		3.49	Single	0	500, 1000, 1500
		4.53	Single	0	500, 1000, 1500
2	Two	2.68	Double	30	1500
		2.68	Double	60	1500
		2.68	Double	90	1500

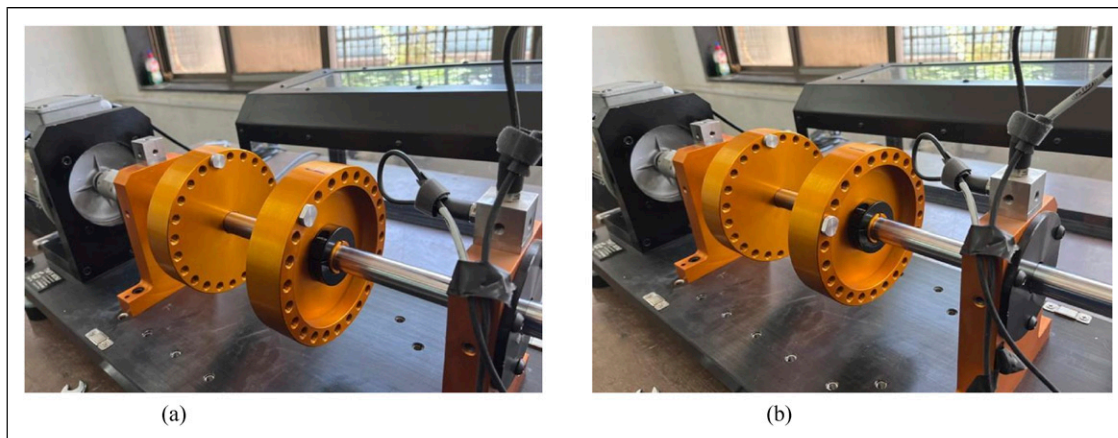


Figure 4. Rotating unbalance in multiple planes (a) 30 degrees (b) 60 degrees.

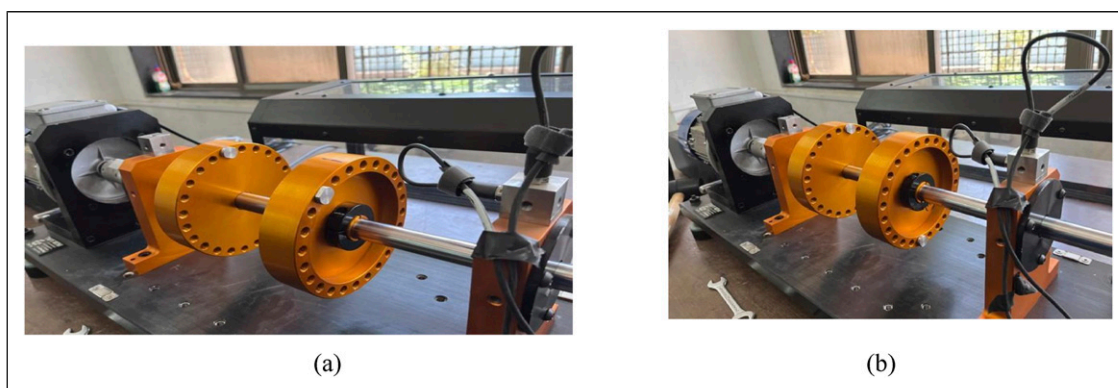


Figure 5. Rotating unbalance in multiple plane (a) 30 degrees (b) 180 degrees

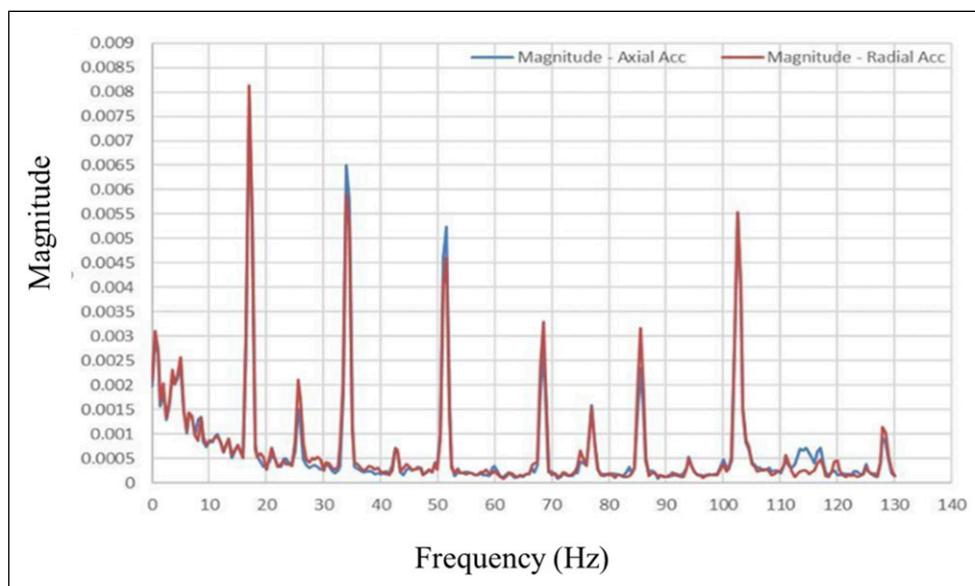


Figure 6. Variation of acceleration for healthy system.

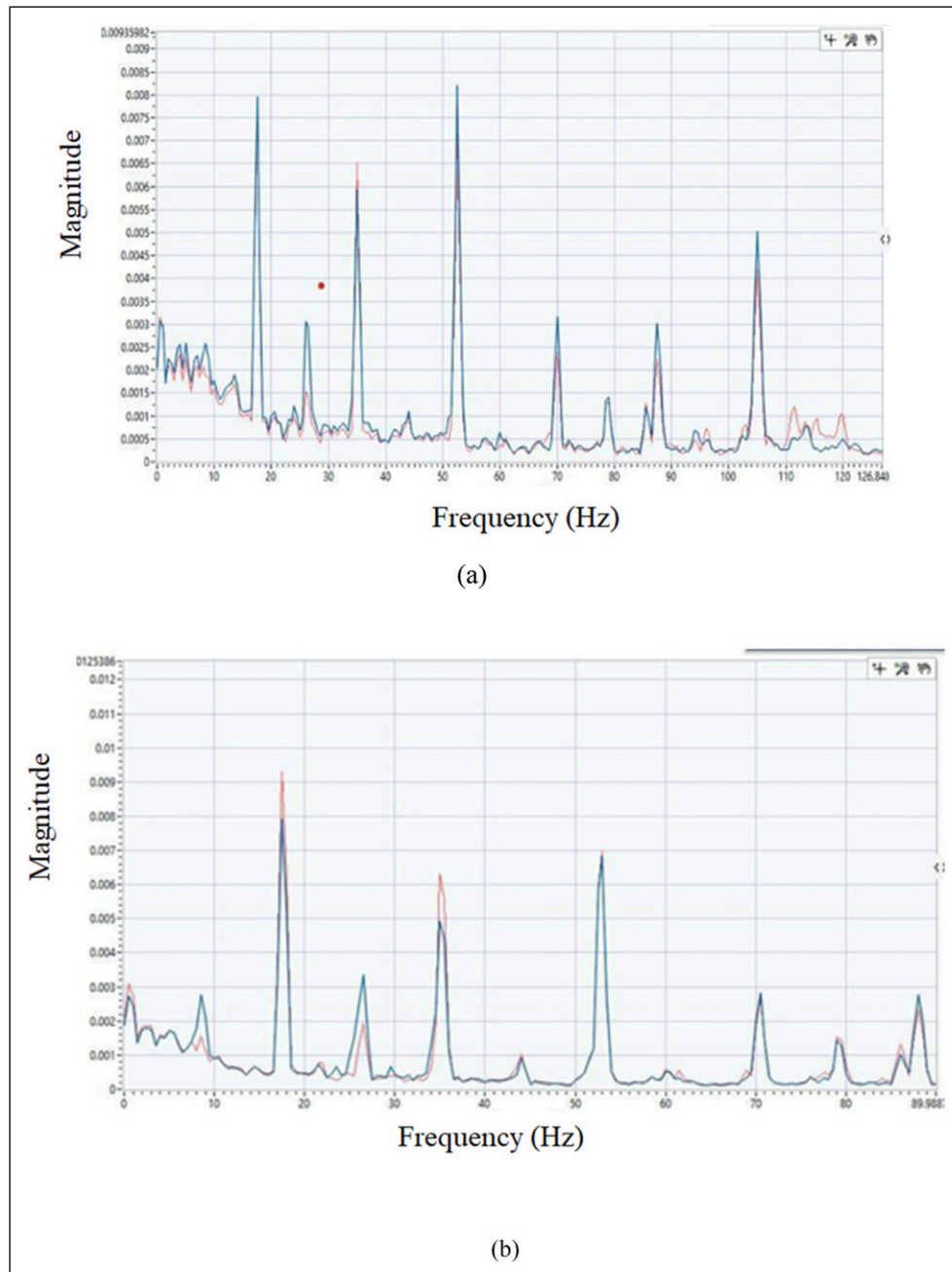


Figure 7. Variation of acceleration with (a) unbalance mass of 2.68 g (b) unbalance mass of 3.49 g rotating at 500 rpm.

which in turn will minimize production downtime. Monitoring time and frequency signals in rotating machines helps assess the condition of all components. Early fault detection prevents major damage, lowers maintenance costs, and boosts productivity,^{20,21}

Experimental set up

Experimental setup in Figure 2 shows that a shaft of 12 mm diameter turned grounded and polished stainless steel shaft is supported on two bearing blocks at ends. Specification of

fault simulator used experimentation are as follows.¹⁹ Three phase induction motor with rating of 0.25 HP with VFD drive for speed control, Maximum speed of 3000RPM, Two bearing blocks (aluminum alloy) with drilled holes for varying the rotor span Deep groove double row concealed bearings, Two Anodized Aluminum discs (150 mm dia with 30 mm thick) with evenly spaced holes at the sides for adding weights for simulating and correcting un-balance. EDM slots with clamping screws are provided to add fins (3 nos) to facilitate blade-pass frequency and blade rub studies. Mounting pads for the magnetic base at

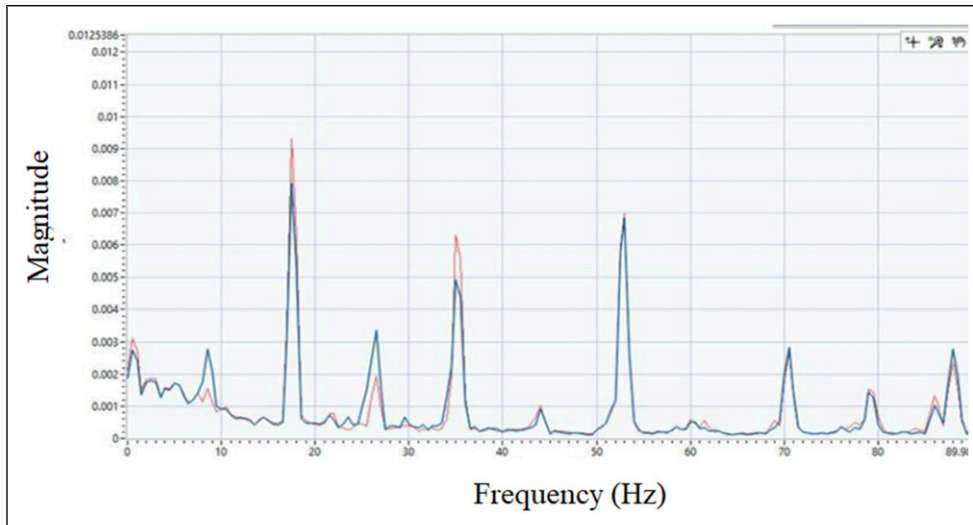


Figure 8. Variation of acceleration with unbalance mass of 4.53 g rotating at 500 rpm.

bearing and motor housing. Slotted plates for placing tachometer and proximity probes 200*500*20 mm, powder-coated aluminum-alloy base plate. Alternate holes are provided for adjusting the effective distance between bearing mounts. Proper damping pads are provided for isolation. Stiffening is provided to avoid resonances. Stringed/rubberized supports to resist excessive vibrations during whirling.

Experimental Methodology

To simulate a fault due to unbalance, multiple holes are drilled on the disc in a circular array in which tiny masses can be mounted to achieve the effect due to out-of-plane and in-plane unbalance. The attachment of masses in particular holes gives the provision to achieve out of plane or in plane unbalanced condition. The experiment was conducted on the Tiera MFS, considering different variables. The fault diagnosis procedure involved the following steps.

- (i) The rotating disk was first tested at various speeds without any added weights to observe its normal (healthy) behavior.
- (ii) It was then tested at the same speeds with different weights attached to one side.
- (iii) Next, the disk was run at different speeds with the same weight fixed on one side.
- (iv) To analyze different imbalance types—static, couple, and dynamic—multiple weights were placed on one side at varying angles.
- (v) Additional tests involved placing multiple weights at different positions, angles, and planes.

These tests provided diverse data to examine how changes in weight, speed, and angle affect vibration patterns. The results were immediately analyzed using T-Vib

software, focusing on time and frequency-based vibration graphs. Initially vibration spectrum were taken for a rotating disk without any unbalance mass. To understand the effect of various rotating unbalance masses in a single plane on vibration, the vibration spectrum was captured at various speeds as 500, 1000, 1500 rpm. The unbalance masses (in grams) varied from 1.4g, 1.92g, 2.68g, 3.49g, 4.53g. To study the effect of rotating unbalanced masses positioned at different angles in a single plane as shown in Figure 2. An experiments were conducted using two masses of 2.68g each, mounted on a rotating disk in the same plane at the following angular separations.

- (i) 30 degrees apart
- (ii) 60 degrees apart (Figure 3(a))
- (ii) 90 degrees apart (Figure 3(b))

Table 1 shows the data for trials i.e masses added, their plan, position in which plane and speeds. On similar vein, the masses now were added in two different planes separated as shown in Figures 4 and 5. The experiment was repeated for the above case. It is very clear from above that the effect of the rotating unbalance mass in single and multiple planes respectively as shown in Figures 2 – 3 and Figures 4 – 5 is extensively studied at various speeds. The next section is detailed about the result and discussion.

Results and discussion

The goal of this experimental study was to investigate the influence of rotating unbalanced masses—at varying rotational speeds and in both single and multiple planes—on the vibration spectrum. The experiments were conducted using the Tiera Machinery Fault Simulator. Additionally, the effect of positioning unbalanced rotating masses at different angular locations was analyzed. The initial phase

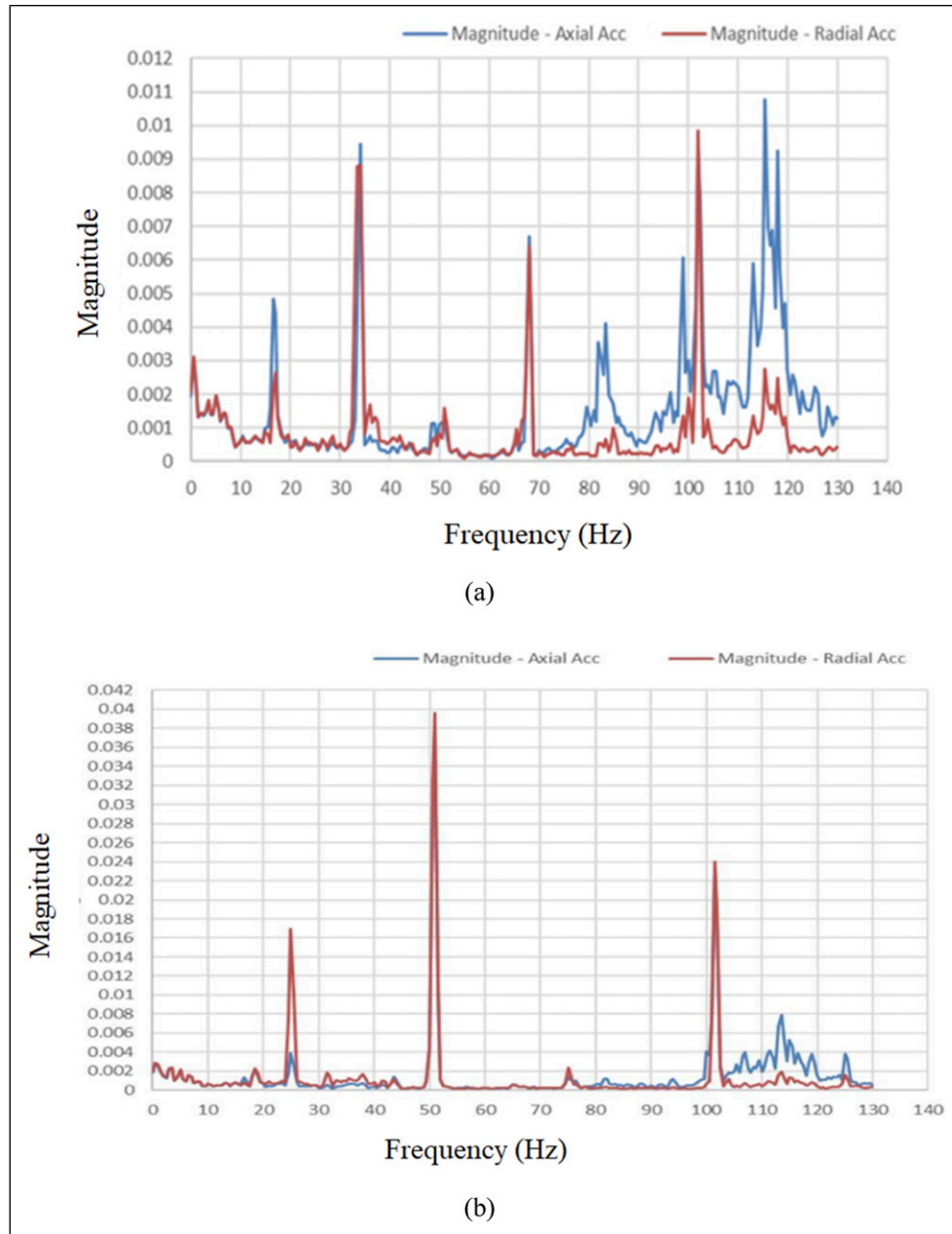


Figure 9. Variation of acceleration with (a) unbalance mass of 1.4 g at 1000 rpm (b) unbalance mass of 1.4 g rotating at 1500 rpm.

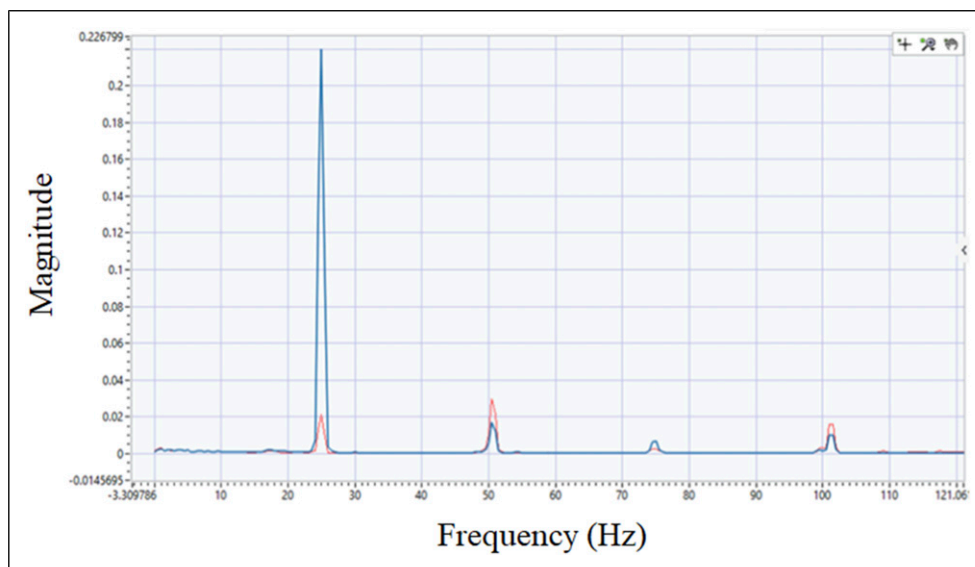
of the study focused on single-plane unbalance, as illustrated in Figures 2 and 3, which depict the experimental setup. Figure 2(b) shows a small mass affixed to the rotating disc, while Figure 2(c) indicates the zero-degree reference marking on the rotor. The unbalanced mass was rotated at different speeds, and the resulting vibrations were recorded over time using a vibration sensor.

The acquired vibration data was analyzed using **T VibLab** software, employing the Fast Fourier Transform (FFT) technique to detect any abnormalities in the machine's behavior. This analysis process was repeated for different unbalanced mass configurations. The

impact of increasing the rotating unbalance mass on the frequency-domain spectrum is illustrated in Figure 6 corresponding to the healthy condition. An increase in the amplitude of radial acceleration at specific frequency peaks is clearly observed. Further results for unbalance masses of 2.68 g, 3.49 g, and 4.53 g at a constant speed of 500 RPM are presented in Figures 7(a) and (b), and 8(a), respectively. Similar experimental procedures were also carried out at rotational speeds of 1000 RPM and 1500 RPM. The amplitude-versus-frequency spectra illustrate both radial and axial acceleration components, with the

Table 2. Axial acceleration and Radial acceleration.

Speed (RPM)	Mass (grams)	Axial acceleration (m/s^2)	Radial Acceleration (m/s^2)
500	No mass	0.000992773	0.00114488
500	1.4	0.00133713	0.00133241
500	1.92	0.00257288	0.0018868
500	2.68	0.00412812	0.00390361
500	3.49	0.00290356	0.00492729
500	4.53	0.00265987	0.00141504
1000	No mass	0.00212036	0.00156565
1000	1.4	0.00482822	0.00266017
1000	1.92	0.00150873	0.0019784
1000	2.68	0.00132759	0.00305698
1000	3.49	0.00549224	0.00761732
1000	4.53	0.0229331	0.0192736
1500	No mass	0.00979274	0.00527955
1500	1.4	0.00393124	0.01696
1500	1.92	0.132503	0.014664
1500	2.68	0.125716	0.013262
1500	3.49	0.192372	0.019323
1500	4.53	0.229816	0.022823

**Figure 10.** Variation of acceleration with two masses of 2.6 g attached at angle of 30 degree to each other rotating at 1500 rpm.

radial component represented in blue and the axial component in red, as shown in Figures 6 and 7.

The experimental results clearly show that changes in the unbalanced mass in a single plane greatly affect vibration, as the increase in mass produces higher centrifugal force, which in turn amplifies the vibration response of the system (Figure 8 and 9). Table 2 provides the peak values for both axial and radial acceleration at different speeds and unbalanced masses. The data shows that as the speed and the unbalanced mass increase, the radial acceleration also

increases noticeably. To study the effect of rotating unbalance multiple masses on a single plane, we selected 2.6g mass rotating at 1500 rpm by varying the angle between them i.e. 30, 60, 90 and 180 degrees. Figures 10–12 show the frequency spectrum measured experimentally. The variation of acceleration with change in angle between two masses is shown in Table 3. It shows that acceleration increases with increase in angle upto 60 degrees but then it decreases after that because of partial cancellation and system dynamic effects.

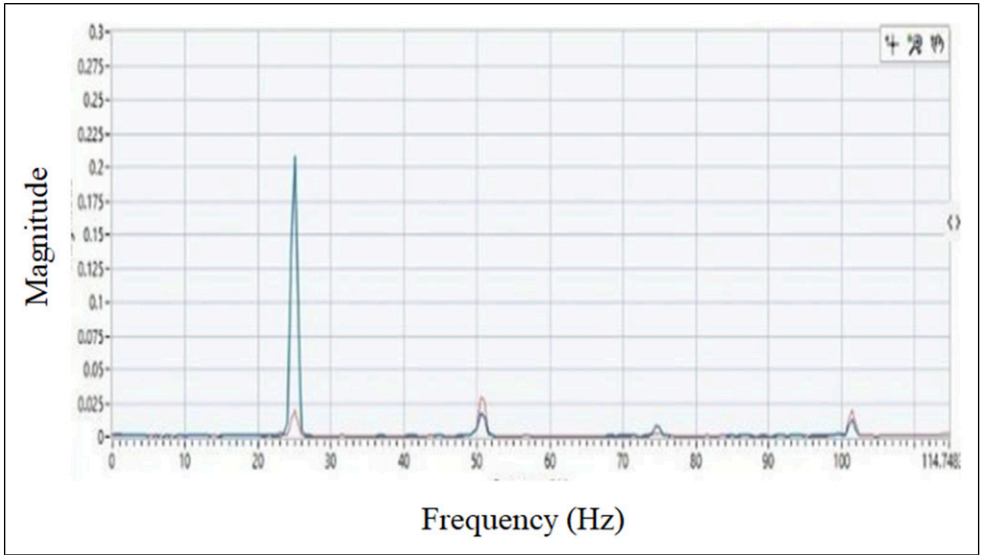


Figure 11. Variation of acceleration with two mass of 2.6 g attached at angle of 60 degree to each other rotating at 1500 rpm.

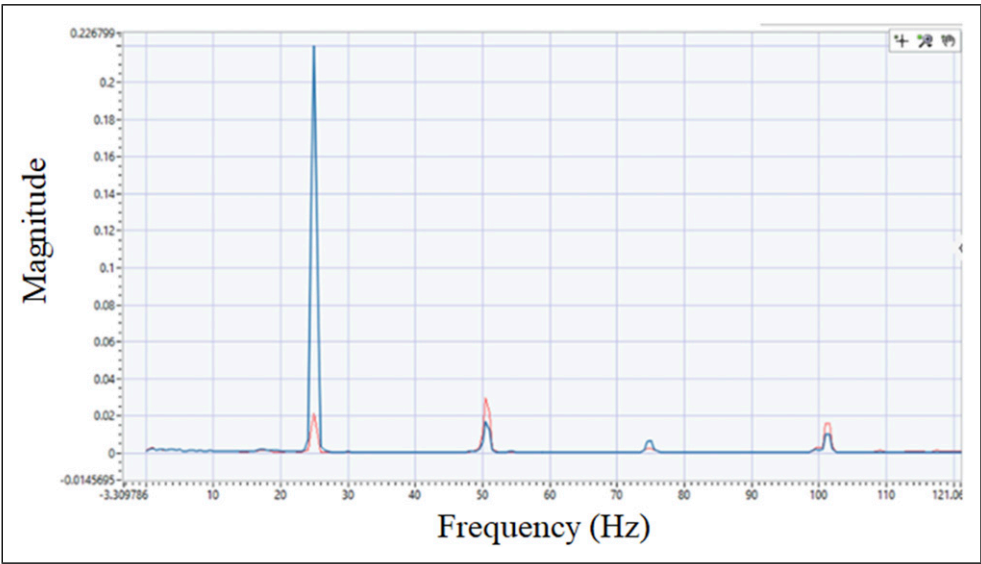


Figure 12. Variation of acceleration with two masses of 2.6 g attached at angle of 30 degree to each other rotating at 1500 rpm.

Table 3. Axial and radial acceleration.

Speed (RPM)	Angle between two masses (degrees)	Axial acceleration (m/s ²)	Radial Acceleration (m/s ²)
1500	Healthy	0.009793	0.005
1500	0°	0.125716	0.01
1500	60°	0.210424	0.01
1500	90°	0.190307	0.01
1500	180°	0.02362	0.005

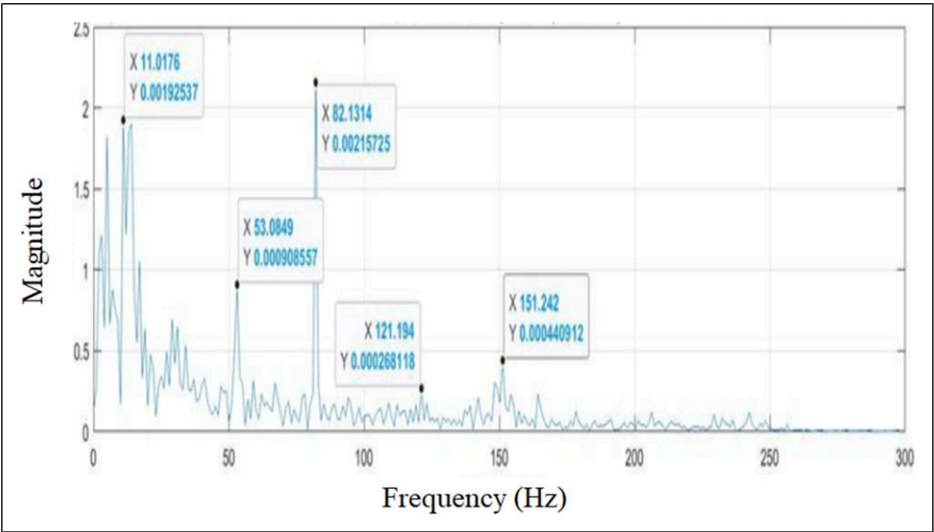


Figure 13. Amplitude versus frequency for shaft with healthy condition rotating at speed of 800 rpm mentioned as Case 0.

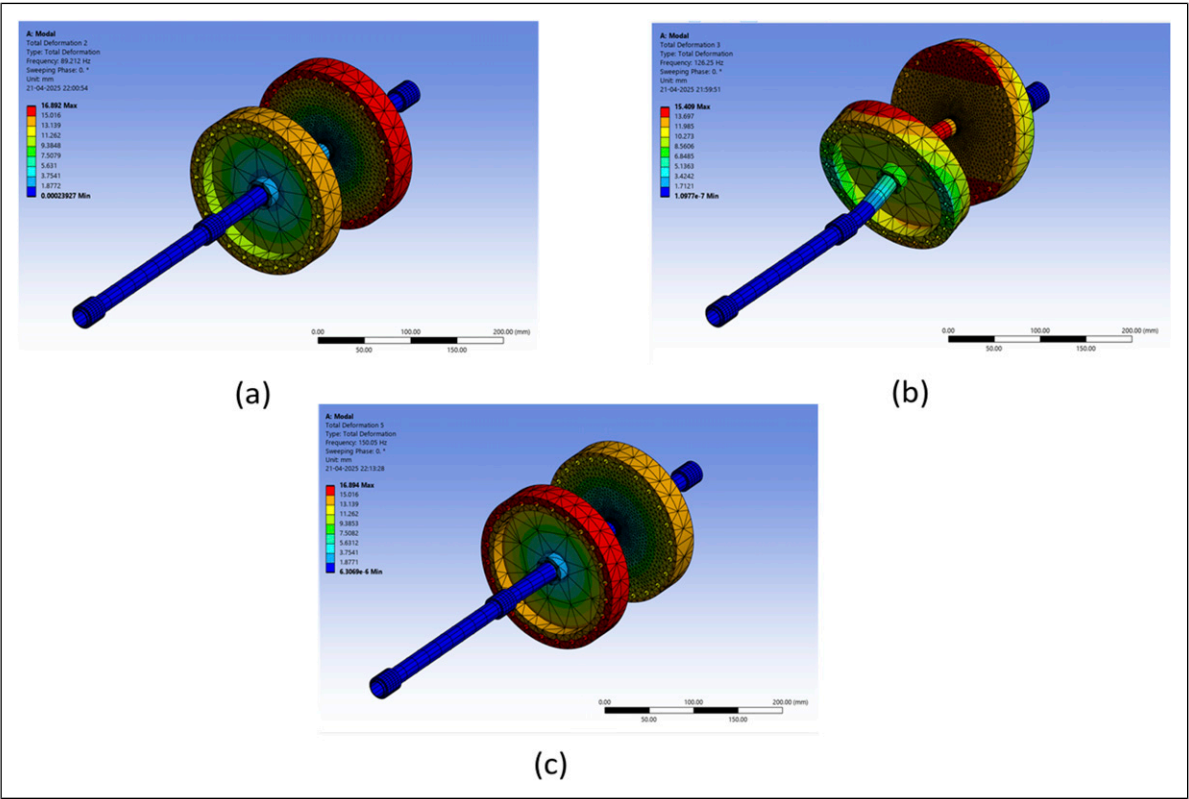


Figure 14. 3D FEM Results for healthy condition Case 0.

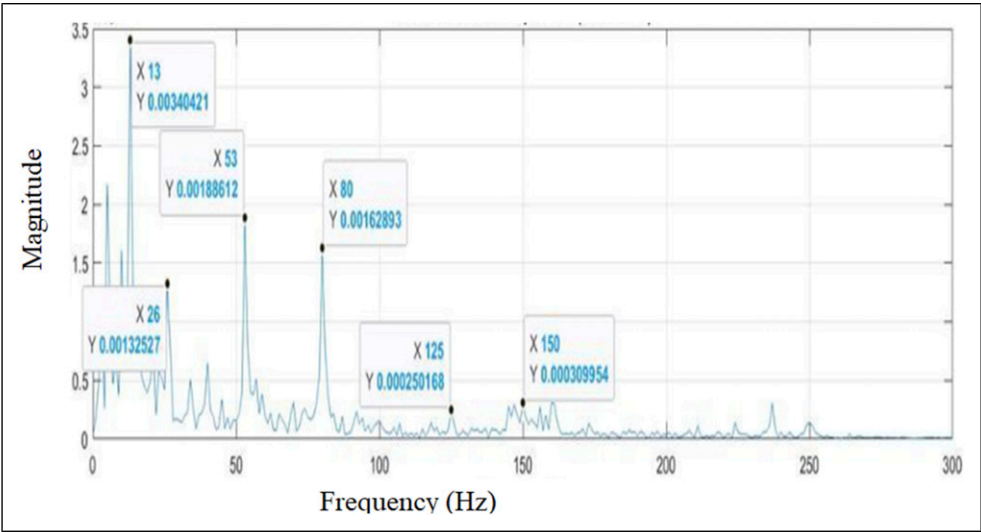


Figure 15. Amplitude versus frequency for shaft with unbalanced mass of 1.4 g rotating at speed of 800 rpm mentioned as Case I.

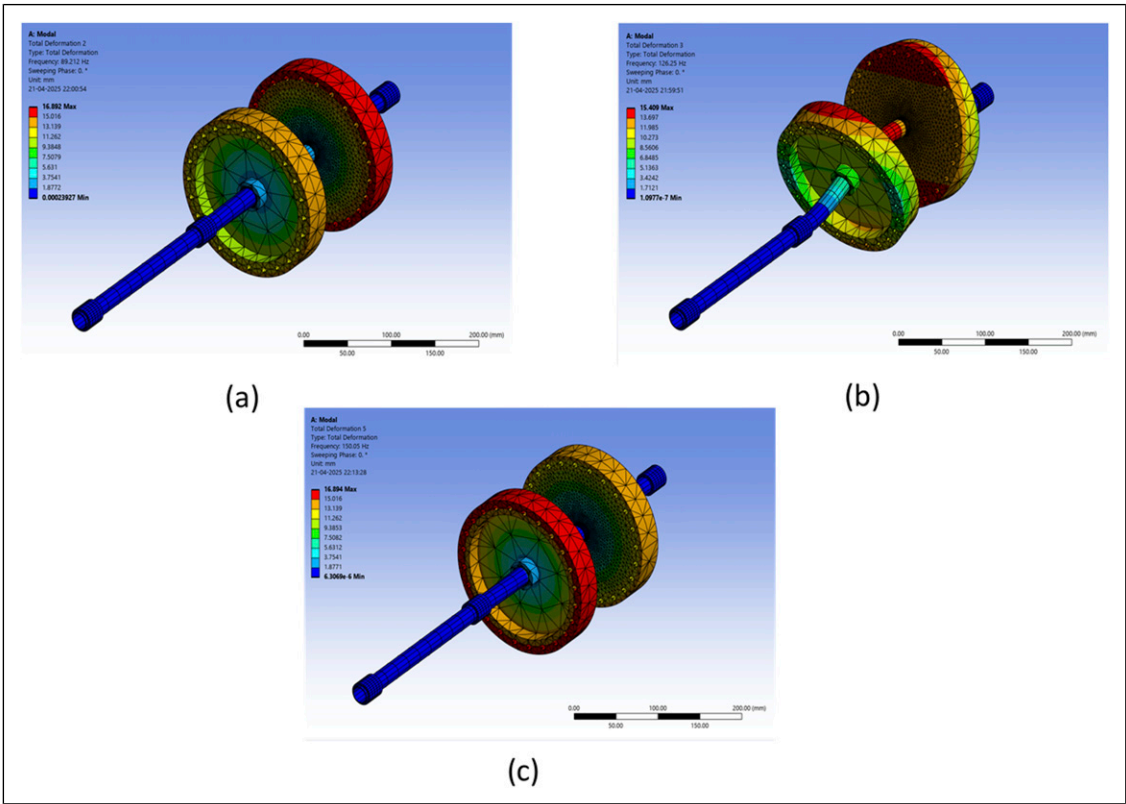


Figure 16. 3D FEM Results rotor with unbalance mass of 1.4 g rotating at speed of 800 rpm referred as Case I.

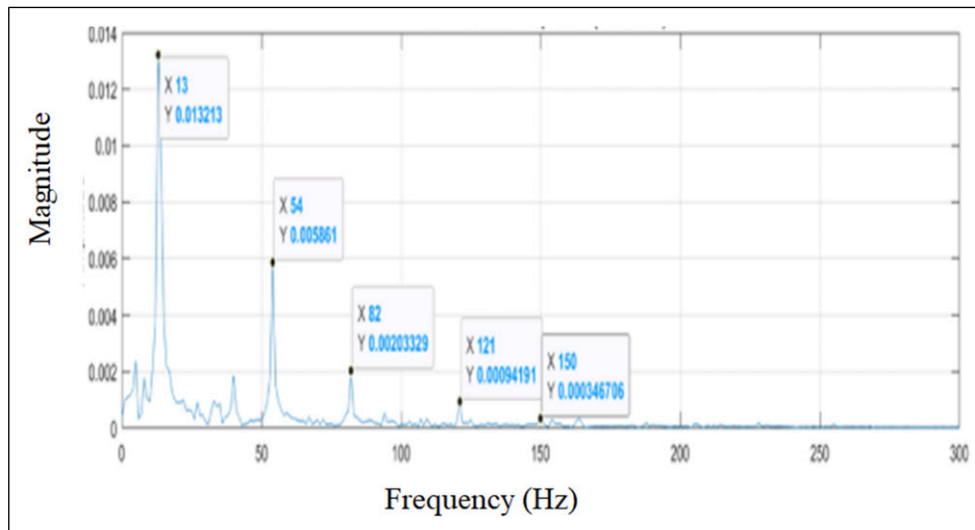


Figure 17. Amplitude versus frequency for shaft with unbalanced masses of 1.4 g, 1.96 g, and 3.47 g rotating at speed of 800 rpm mentioned as Case 2.

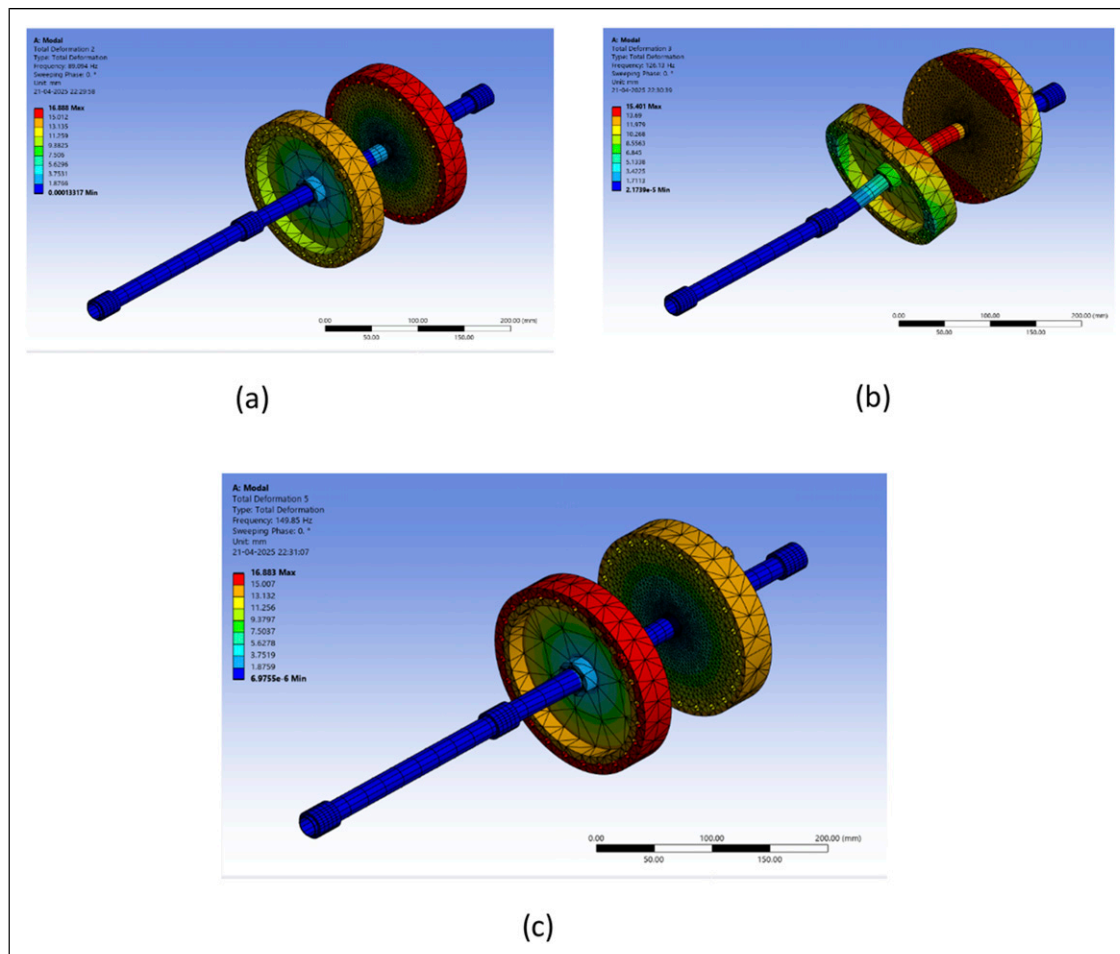


Figure 18. 3D FEM Results rotor with unbalance masses of 1.4 g, 1.96 g, and 3.47 g rotating at speed of 800 rpm referred as Case 2.

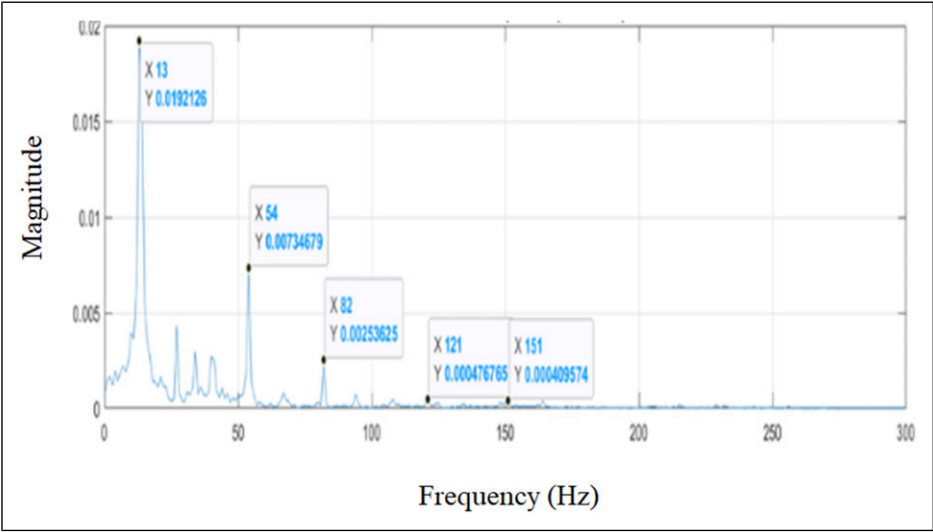


Figure 19. Amplitude versus frequency for shaft with unbalanced masses of 1.4 g, 1.96 g, 3.47 g, and 4.96 g rotating at speed of 800 rpm mentioned as Case 3.

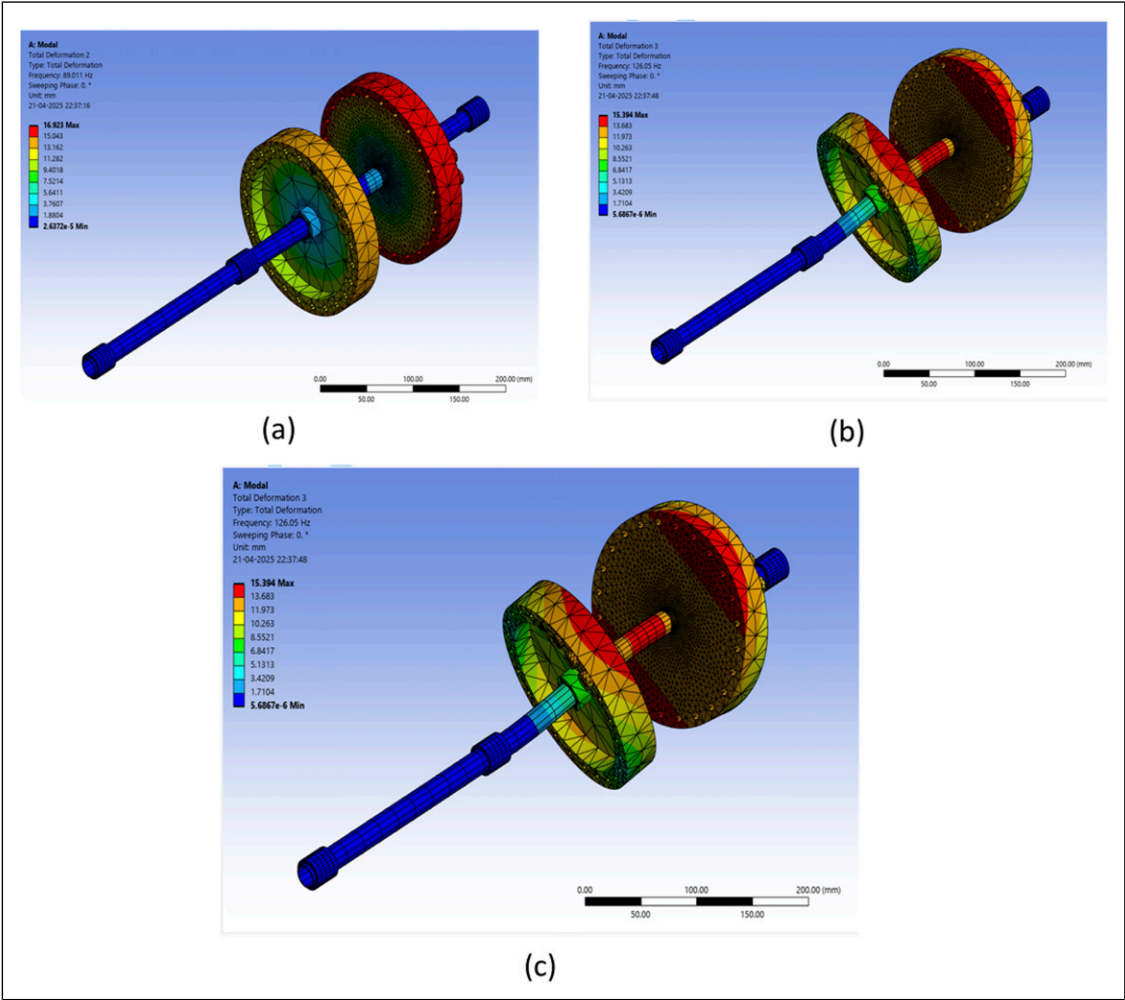


Figure 20. 3D FEM Results rotor with unbalance masses of 1.4 g, 1.96 g, and 3.47 g rotating at speed of 800 rpm referred as Case 3.

Table 4. Result comparison table.

Sr.No	Case	Experiment Result (Hz)	3D FEM Result (Hz)	Percentage Deviation
1	Case 0 (Healthy)	82.1314 Hz	89.21 Hz	8.60%
		121.194 Hz	126.25 Hz	4.17%
		151.242 Hz	150.05 Hz	0.79%
2	Case 1	80 Hz	89.205 Hz	11.51%
		125 Hz	126.24 Hz	0.99%
		150 Hz	150.03 Hz	0.02%
4	Case 2	82 Hz	89.094 Hz	8.65%
		121 Hz	126.13 Hz	4.24%
		150 Hz	149.85 Hz	0.10%
5	Case 3	82 Hz	89.011 Hz	8.55%
		121 Hz	126.05 Hz	4.17%
		151 Hz	149.71 Hz	0.85%

On Similar vein the effect of rotating unbalance on vibration was studied in multiple planes. To start with two single masses of 2.6 g were attached to two rotors rotating in different planes at 500 rpm. The time response was recorded which was then converted to frequency response through FFT processing in T-Viblab software. The frequency response recorded by machinery fault simulator as shown in Figure 10. Then the same two masses were attached on two different rotating discs but at 30 degrees to each other. The time domain and then frequency domain spectrum were recorded at 1000 and 1500 rpm. Same procedure was repeated for other rotating unbalance masses as mentioned above.

The corresponding material properties for the tiny aluminum masses were taken as Modulus of Elasticity $E = 70$ GPA, Poisson's ratio $\mu = 0.28$, and density $\rho = 2710 \text{ kg/m}^3$, respectively. The rotational velocity of 83.77 rad/sec was given to the shaft in ANSYS. A simply supported boundary condition providing only axial movement was imparted to both ends of the shaft. The experimental results from Tiara simulator are obtained in the form of amplitude versus time. This data was further processed through FFT in MATLAB to get it in the form of amplitude versus frequency, from which dominant peaks of frequencies are captured.

Validation through Finite Element analysis of rotating unbalance

To validate the experimental results with 3D FEM, we modeled a shaft with two discs on it in SolidWorks. We also modeled aluminum masses with different weights to simulate unbalance. A modal analysis was performed for 800 RPM speed in ANSYS.²² Young's modulus, Poisson's ratio and steel shaft density were 200 GPA, 0.3 and 7850 kg/m^3 , respectively. In the healthy condition (case 0), where no unbalanced mass was added, the experimental results showed frequency peaks at around 82 Hz, 121 Hz and 151 Hz as shown in Figure 13. In the 3D FE simulation, three natural frequencies were identified as 89.21 Hz, 126.25 Hz, and 150.05 Hz as shown in

Figure 14. In Figure 14 it is seen that (a), (b) and (c) are torsion-dominant coupled modes.

In Case 1, only one unbalanced mass of 1.4 g was placed on a disc. The experimental results showed three frequency peaks at around 80 Hz, 125 Hz, and 150 Hz as shown in Figure 15. From the 3D FE simulation as shown in Figure 16 the natural frequencies were found to be 89.205 Hz, 126.24 Hz, and 150.03 Hz. Overall, the comparison of experimental investigation and 3D FEM model predictions for natural frequencies shows good agreements with deviations below 10%. In Case 2, three masses were placed on each disc (1.4 g, 1.96 g, and 3.47 g). (Figure 17 and 18).

The experimental results are 82 Hz, 121 Hz and 150 Hz. The 3D FE results showed 89.094 Hz, 126.13 Hz and 149.85 Hz. It is seen from comparison of experimental and 3D FE simulation results taken for case 2 unbalance are in good agreement. In Case 3, four masses (i.e 1.4g, 1.96g, 3.47g, and 4.96g) were added to each disc. This case had the highest level of out-of-plane unbalance. The experimental results showed frequency peaks at 82 Hz, 121 Hz, and 151 Hz. However, in comparison to this, the 3D FEM results obtained from ANSYS give the three torsion dominant frequencies as 89.011 Hz, 126.05 Hz and 149.71 Hz as shown in Figure 19 and 20. The close observation of the results of frequencies from experiments and computation show acceptable validation with minimum deviation as shown in Table 4.

Conclusion

This experimental study investigates various aspects of machinery performance and vibration analysis through simulations using the MFS. The study focuses on the effect of rotating unbalanced mass in single and multiple planes on vibrations at various speeds. The analysis of the frequency response plots clearly indicates a significant difference between the behavior of a balanced system and that of an unbalanced one. In the balanced condition, the system remains relatively stable, and the frequency spectrum does not display any

noticeable or sharp peak at the excitation frequency. This implies that the machine is operating smoothly without additional dynamic forces acting on it. In contrast, when the system is subjected to unbalanced conditions, distinct and dominant peaks emerge in the frequency plot precisely at the excitation frequency of the machine. These peaks highlight the presence of periodic forces induced by the unbalance. Furthermore, as the magnitude of the unbalanced mass is increased, or as the rotational speed of the system rises, the vibration amplitude also grows significantly. This increase can be attributed to the larger centrifugal forces generated during rotation, which exert greater dynamic loading on the system. Consequently, the severity of vibrations becomes more pronounced with higher levels of unbalance and speed, indicating a direct relationship between unbalance, rotational speed, and vibration response.

In summary, the position of the unbalanced weight can significantly impact the amplitude of vibration, and understanding this can help to improve machine performance and longevity. It is observed from experimental and 3D FEM results that the unbalance fault in multi-plane is dominantly influencing the amplitude of vibration. This work can be extended by combining multiple faults in rotating machines to understand the performance of a machine as a future work.

Declaration of conflicting interests

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